



FITNESS & WELLNESS MOBILE APP FOR MEMBER SELF-SERVICE

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Database Systems Engineering and Distributed Backend Development

ABSTRACT

- Smart, user-friendly mobile solution for centralized fitness and wellness self-service.
- Supports registration, workout tracking, exercise plans, class/trainer booking, membership management, progress monitoring, notifications and wellness guidance.
- Outcome: greater convenience, personalization, engagement and reduced need for manual assistance.

PROBLEM STATEMENT & OBJECTIVES

- Problem: Fitness services can require manual assistance and fragmented access to schedules, memberships and wellness information.
- Objective 1: Provide a single digital platform for member self-service.
- Objective 2: Enable personalized fitness, booking, membership and progress features.
- Objective 3: Improve convenience, engagement and access to wellness services.
- Scope: Member registration, fitness tracking, plans, bookings, memberships, progress, notifications and wellness guidance.

EXISTING VS PROPOSED SYSTEM

Feature	Existing / Manual	Proposed App
Member access	Multiple/manual touchpoints	Single digital platform
Booking	Assisted scheduling	Class & trainer booking
Membership	Manual support	Self-service management
Progress & wellness	Limited centralization	Tracking + guidance + notifications

FEASIBILITY STUDY

- Technical: React.js, Node.js/Express.js, REST APIs, MySQL, Sequelize, JWT, Twilio and Git/GitHub are identified in the project plan.
- Economic: Uses widely available development tools and open-source technologies.
- Operational: Centralized self-service access is designed to reduce manual assistance and simplify member interactions.
- Schedule: Academic PBL project for Y25 – 2026–2027 Trimester 4.

RESULTS & TESTING

- Centralized member services
- CRUD + validation + API integration
- JWT-secured access and role control

CONCLUSION

- A unified self-service platform is defined around the project objectives.
- The design connects fitness, membership and wellness services in one flow.

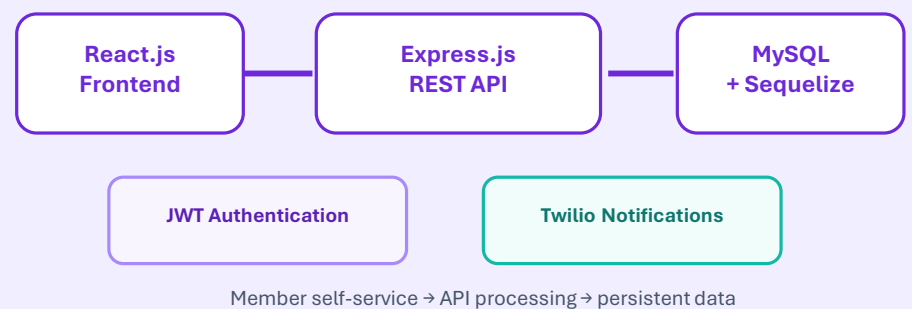
FUTURE SCOPE

- Personalized recommendations and analytics
- Expanded notifications and wellness guidance
- Scalable deployment and additional integrations

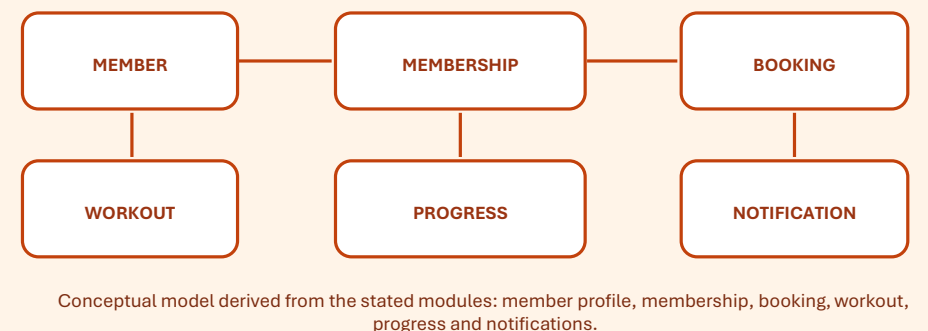
TOOLS & TECHNOLOGIES

- Frontend: React.js
- Backend: Node.js + Express.js | RESTful APIs
- Database: MySQL (RDBMS) | ORM: Sequelize
- Authentication & Security: JWT + role-based access control
- Communication: Twilio | Development: VS Code, Git, GitHub | Docker if required

SYSTEM ARCHITECTURE



ER DIAGRAM & DATABASE DESIGN



IMPLEMENTATION / UI MODULE PREVIEWS

